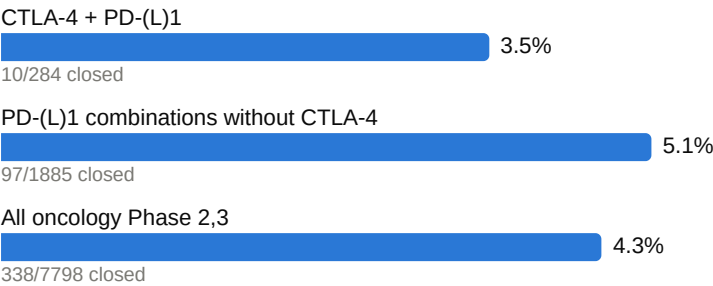


# CTLA-4 + PD-(L)1: 3 molecules stopped early, all monoclonal antibodies

10 stopped trials, but 3 molecules — all monoclonal antibodies: Ipilimumab, Erfonrilimab and Tremelimumab. Against 5.1% for PD-(L)1 combinations without CTLA-4 and 4.3% across all oncology Phase 2,3 trials in the same window. Rates count trials that stopped early for efficacy, safety or benefit–risk reasons; trials that completed and missed their endpoints are not counted.

|                                            |                                       |                                                          |                                                                                                                  |
|--------------------------------------------|---------------------------------------|----------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| 3.5%                                       | 9 / 1 / 0                             | 39.6%                                                    | 1.2%                                                                                                             |
| 10 of 284 closed trials (95% CI 1.9%–6.4%) | efficacy / safety / both (adds to 10) | of 718 trials have closed · 434 still open or unresolved | stopped for a biological reason within 3 years of starting (95% CI 0.5%–2.6%; 380 still at risk; 1.5% area-wide) |

## How this compares



**What is in this class:** 276 distinct experimental drugs across 718 trials (antibody 714, small molecule 244, unknown 78, adc 32). Most tested: IPILIMUMAB (321), NIVOLUMAB (295), CADONILIMAB (139), DURVALUMAB (129), TREMELIMUMAB (122). A class groups drugs by what they act on; check that the grouping is one you would make before reading the rate as a property of the mechanism.

**How much does this rest on one decision?** The 10 stops came from 10 sponsor-asset programmes across 8 sponsors; the largest (Weill Medical College of Cornell University) contributed 1; removing that programme's trials from both sides leaves 3.2%. Screened alongside 83 other segments in this area, it does not survive a 10% false-discovery correction ( $q=1$ , Benjamini-Yekutieli). A further 7 closed trials here were terminated with no cause recorded in the registry; if every one of them were biological the rate would be 6.0%. Read it as a screen worth checking against the underlying trials.

Bars show the share of closed trials stopped for biological reasons. Confidence intervals overlap where sample sizes are small — read the counts, not just the bars.

## By start cohort

| Trials started | Stops | Closed | Rate | 95% CI     |
|----------------|-------|--------|------|------------|
| 2015–2017      | 0     | 130    | 0.0% | 0.0%–2.9%  |
| 2018–2020      | 7     | 106    | 6.6% | 3.2%–13.0% |
| 2021–2024      | 3     | 48     | 6.2% | 2.1%–16.8% |

## Where the stops sit

Efficacy / futility: 9, Safety: 1.  
Sponsors with most stops: Herlev Hospital (2), Jiangsu Alphamab Biopharmaceuticals Co., Ltd (2), M.D. Anderson Cancer Center (1), Weill Medical College of Cornell University (1), University of Washington (1), H. Lee Moffitt Cancer Center and Research Institute (1).

## The stopped trials

| Trial       | Ph | Sponsor                                     | Experimental drugs                       | Stopped | Type     | Registry stop reason                                                                                                                                |
|-------------|----|---------------------------------------------|------------------------------------------|---------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| NCT03274258 | 2  | M.D. Anderson Cancer Center                 | IPILIMUMAB, NIVOLUMAB                    | 2025-11 | Efficacy | The study met its prespecified futility criteria during the prespecified interim futility analysis after enrolling the first cohort of 10 patients. |
| NCT05116917 | 2  | Herlev Hospital                             | NIVOLUMAB, IPILIMUMAB, Influenza vaccine | 2023-10 | Efficacy | Based on the prespecified interim analysis the predictive probability was below 10% for the prespecified endpoint.                                  |
| NCT04925947 | 2  | Weill Medical College of Cornell University | ERFONRILIMAB                             | 2023-05 | Efficacy | Data collected did not support the study endpoints                                                                                                  |

| Trial       | Ph | Sponsor                                      | Experimental drugs                                | Stopped | Type     | Registry stop reason                                                                                                                                         |
|-------------|----|----------------------------------------------|---------------------------------------------------|---------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NCT05001724 | 3  | Jiangsu Alphamab Biopharmaceuticals Co., Ltd | ERFONRILIMAB, LENVATINIB                          | 2023-02 | Efficacy | Because the efficacy data did not meet expectations, the sponsor decided to terminate the study.                                                             |
| NCT03838848 | 2  | Jiangsu Alphamab Biopharmaceuticals Co., Ltd | ERFONRILIMAB, CARBOPLATIN, PEMETREXED, NINGETINIB | 2022-07 | Efficacy | Cohort A,B,C end enrollments. Cohort D and E were considered to have no significant clinical benefit at the SMC meeting and decided to terminate enrollment. |
| NCT03522584 | 2  | University of Washington                     | DURVALUMAB, TREMELIMUMAB                          | 2021-12 | Efficacy | Terminated by the sponsor due to lack of efficacy in study treatment.                                                                                        |

The 6 most recent of 10 stopped trials. Every molecule behind all 10 is named above; the complete trial list, the trials still running and the terminations with no readable cause ship with the dataset and the evidence package.

**Method**

Denominator: ClinicalTrials.gov interventional Phase 2,3 oncology trials started 2015–2024 that have closed (completed or terminated). Numerator: terminated trials whose registry stop reason is classified as biological (efficacy, safety or benefit–risk) by Classification V2 — held-out precision 95.5%, recall 95.3% (n=600). Drugs are linked to ChEMBL and the NCI Thesaurus; 76% of industry oncology trials in this window carry a resolved drug target, and a trial without one cannot enter a mechanism class. Intervals are Wilson 95%.

**Limits**

Not a failure rate: trials that completed with negative results are not counted, and programs discontinued after a completed trial do not appear. Stop reasons are sponsor-reported. This is a closed-trial proportion, not a time-to-event analysis: only 39.6% of trials in this segment have closed, and a trial that stops early enters the denominator sooner than one that runs to completion, which can inflate the rate in immature segments. Recent cohorts have fewer closed trials, so their rates are less stable. Research signals, not clinical or investment advice.

**Any mechanism, sponsor or indication, updated weekly.** The dataset behind this brief covers every stopped oncology trial with an efficacy or safety signal plus the full denominator universe. Free sample and licensing: [clinicaltrialfailures.com/data-licensing](https://clinicaltrialfailures.com/data-licensing)

Sources: ClinicalTrials.gov (NLM); ChEMBL (EMBL-EBI, CC BY-SA 3.0); NCI Thesaurus (NCI); RxNorm/RxClass (NLM); SEC EDGAR. Classification, linkage and rates by Clinical Trial Failures. Rebuilt weekly; this brief covers trials started 2015–2024.